Lecture 9 v2:
An Introduction to Instrumental Variables

25117 - Econometrics

Universitat Pompeu Fabra

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What we learned in the last lesson

- Statistical studies are evaluated by asking whether the analysis is internally and externally valid.
- A study is internally valid if the statistical inferences about causal effects are valid for the population being studied. In regression estimation of causal effects, there are two main types of threats to internal validity:
  - First, OLS estimators are biased and inconsistent if the regressors and error terms are correlated.
  - Second, confidence intervals and hypothesis tests are not valid when the standard errors are incorrect.
- A study is externally valid if its inferences and conclusions can be generalized from the population and setting studied to other populations and settings.

Instrumental Variables

Consider the following regression model:

$$Y_i = \beta_0 + \beta_1 X_i + u_i$$

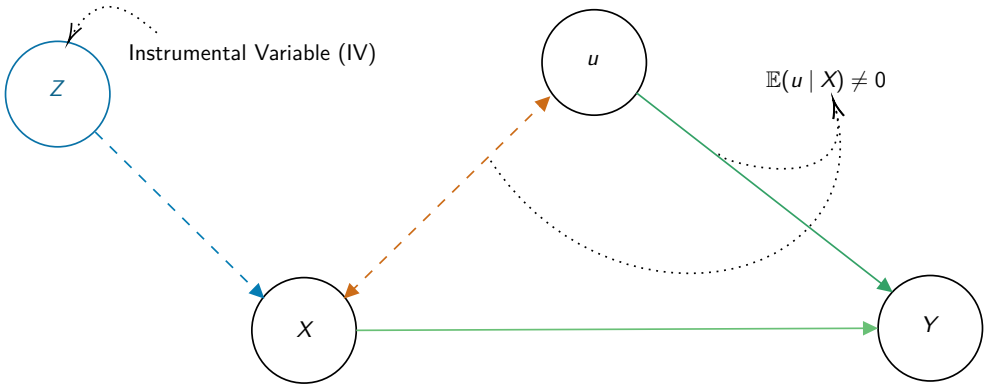
- OVB, Simultaneous causality bias, Errors-in-variables bias... all three result in $\mathbb{E}(u \mid X) \neq 0$
- Instrumental variables regression can eliminate bias when $\mathbb{E}(u \mid X) \neq 0$, using an instrumental variable (IV), Z.
- Intuitively, IV regression breaks X into two parts:
  1. an *endogenous* part that might be correlated with u, and
  2. an *exogenous* part that is not.

  By isolating the part that is not correlated with u, it is possible to estimate β_1

Instrumental Variables

For an instrumental variable (an "instrument") Z to be valid, it must satisfy two conditions:

1. **Instrument relevance:** Z is a strong determinant of X ($corr(Z, X) \neq 0$)
2. **Instrument exogeneity (exclusion restriction):** Z is unrelated to u ($corr(Z, u) = 0$)



Two Stages Least Squares (2SLS)

Assume that Z satifies both conditions. Then the 2SLS estimator of β_1 is obtained by

- **First Stage** — Regress X_i on Z_i (including the intercept), and obtain the predicted values $\hat{X}_i$.
- **Second Stage** — Regress Y_i on $\hat{X}_i$ (including the intercept), the coefficient on $\hat{X}_i$ is the 2SLS estimator $\hat{\beta}_1^{2SLS}$

Formally, suppose that Z satisfies both conditions. As a first step, suppose that we use it as a proxy for X. The reduced-form estimator, β_1^{RF}, reads

$$\beta_1^{RF} = \frac{cov(Z_i, Y_i)}{var(Z_i)} = \frac{cov(Z_i, \beta_0 + \beta_1 X_i + u_i)}{var(Z_i)} = \beta_1 \frac{cov(Z_i, X_i)}{var(Z_i)}$$

$$\Rightarrow \beta_1 = \frac{cov(Z_i, Y_i)}{cov(Z_i, X_i)}$$

The IV estimator (replacing population covariances with the sample covariances) is a consistent estimator of β_1:

$$\hat{\beta}_1^{2SLS} = \frac{s_{ZY}}{s_{ZX}} \xrightarrow{p} \frac{cov(Z_i, Y_i)}{cov(Z_i, X_i)} = \beta_1$$

Note that $\hat{\beta}_1^{OLS} = \frac{s_{XY}}{s_{XX}}$.

Two Stages Least Squares (2SLS)

For another way to see it, consider the system of reduced-form equations:

$$\begin{aligned} X_i &= \pi_0 + \pi_1 Z_i + v_i \\ Y_i &= \gamma_0 + \gamma_1 Z_i + w_i \end{aligned}$$

Because Z is exogenous, Z is uncorrelated with both v_i and w_i.

The idea: An exogenous unit change in Z results in a change in X by π_1 units and a change in Y by γ_1 units.

Thus an exogenous change in X by π_1 is associated with an exogenous change in Y by γ_1 units.

So the effect on Y of an exogenous change in X is $\beta_1^{2SLS} = \gamma_1/\pi_1$

A Buttery Example

IV regressions were first developed in 1928 by Philip G. Wright to estimate supply and demand elasticities for agricultural goods,in particular, butter.

Consider the following demand equation:

$$ln(Q_i^{butter}) = \beta_0 + \beta_1 ln(P_i^{butter}) + u_i$$

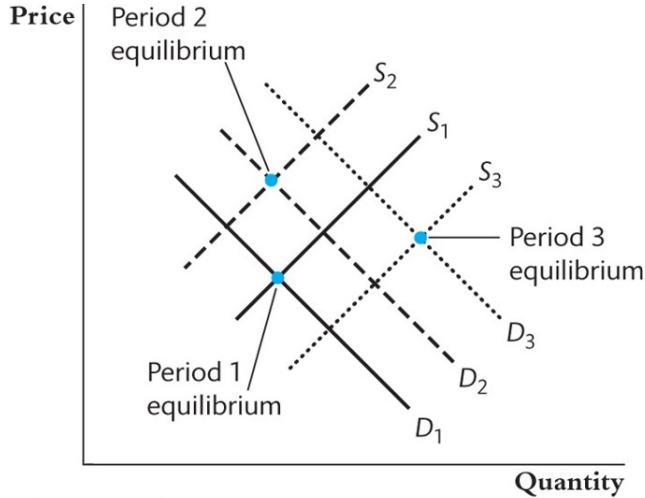
Here, β_1 captures the price-elasticity of butter, i.e., the percentage change in quantities due to a 1% change in price.

Problem: The OLS regression of $ln(Q_i^{butter})$ on $ln(P_i^{butter})$ suffers from simultaneity (reverse causality) bias.

Because prices, $ln(P_i^{butter})$, are simultaneously determined in equilibrium by the interplay of supply and demand, β_1 was unlikely to capture neither the supply or demand elasticity of butter.

A Buttery Example

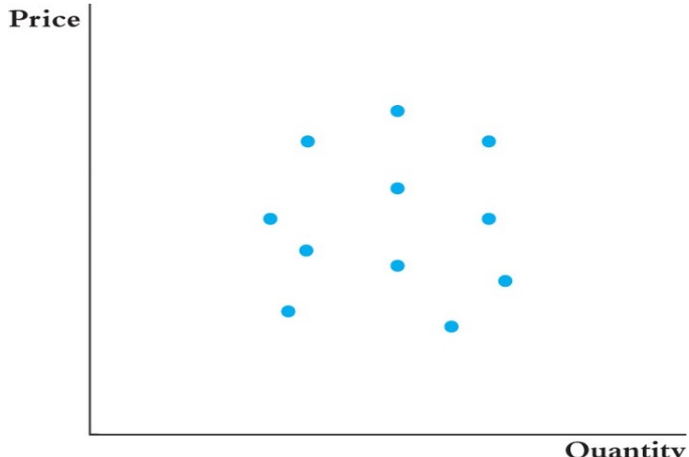
- In every period, prices are determined in equilibrium by the simultaneous movements of supply and demand
- The equilibrium is determined by the intersection of the demand curve D_k and supply curve S_k



(a) Demand and supply in three time periods

A Buttery Example

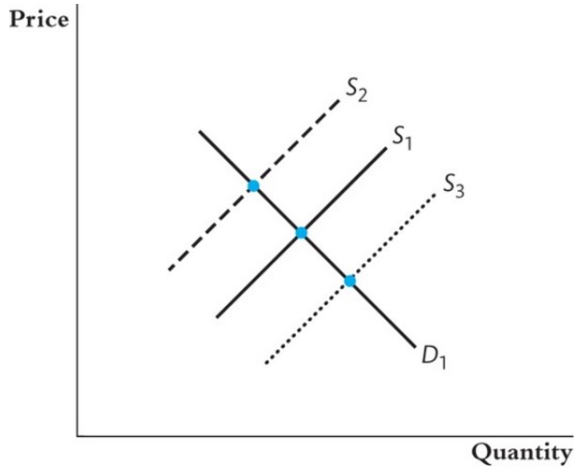
- As a result, the set of $\{Price_k, Quantity_k\}$ is not informative about the specific demand and supply elasticities (i.e., the curves' slopes)
- In this example, demand and supply elasticities could be anything, from perfectly elastic (a small change in P is associated with a high change in Q) to completely inelastic (a high change in P is associated with a little change in Q).



(b) Equilibrium price and quantity for 11 time periods

A Buttery Example

- To determine the elasticity of demand, we need an IV that would shift only supply curves while leaving demand curves unchanged.
- What could be such an instrument?



(c) Equilibrium price and quantity when only the supply curve shifts

A Buttery Example

A potential IV for the butter prices to estimate the demand elasticity for butter could be drought shocks to dairy regions (Z).

It is a plausibly revelant IV: below-average rainfall in a dairy region (Z) could impair grazing and thus reduce butter production at a given price, shifting the supply curve to the left and increasing the equilibrium price (X)

Drought shocks are also plausibly unrelated to the quantities of butter demanded (Y)... except through the price increase (X)!

Can you think of an instrument to estimate the supply elasticity of butter? In the California Schools example, what could be an instrument for the share of subsidized meals?

Large-Sample properties

- As $n \to \infty$, $\hat{\beta}_1^{2SLS} \sim \mathcal{N}(\beta_1, \sigma^2_{\hat{\beta}_1})$ where $\sigma^2_{\hat{\beta}_1} = \frac{1}{n} \frac{var(Zi - \mu_Z)u_i}{[cov(Z_i, X_i)]^2}$
- Statistical inference proceeds in the usual way.
- The justification is (as usual) based on large samples
- This all assumes that the instruments are valid – we'll discuss what happens if they aren't valid shortly.
- However, the OLS standard errors from the second stage regression aren't right because they do not account for the estimation in the first stage
- Instead, use a single specialized command that computes the 2SLS estimator and the correct SEs.

The General IV Regression Model

We can extend the simple IV framework to multiple endogenous regressors (Xs), exogenous regressors or control variables (Ws), and exogenous instruments (Zs).

We say the coefficients $\beta_1, \ldots, \beta_k$ are

- **underidentified** if there are more regressors than instruments ($m < k$).
- **exactly identified** if there are as many regressors as instruments ($m = k$).
- **overidentified** if there are less regressors than instruments ($m > k$).

E.g., with a single endogenous regressor:

$$Y_i = \beta_0 + \beta_1 X_{1i} + \beta_2 W_{1i} + \cdots + \beta_{r+1} W_{ri} + u_i$$

and m instruments, $Z_{1i}, \ldots, Z_{mi}$.

2SLS with a single endogenous regressor

With a single endogenous regressor:

$$Y_i = \beta_0 + \beta_1 X_{1i} + \beta_2 W_{1i} + \cdots + \beta_{r+1} W_{ri} + u_i$$

and m instruments, $Z_{1i}, \ldots, Z_{mi}$.

First Stage:

- Regress X_1 on *all* exogenous regressors, $W_1, \ldots, W_r, Z_{1i}, \ldots, Z_{mi}$, plus intercept.
- Compute predicted values of $\hat{X}_1$

Second Stage:

- Regress Y on $\hat{X}_1$, $W_1, \ldots, W_r$ and the intercept.
- The coefficients from this second stage regression are the 2SLS estimators, but SEs are wrong
- To get correct SEs, do this in a single step in your regression software

Example with $m = 1$ – Price Elasticity of cigarettes

First stage: dependent variable `ltavgprs`

Regressor	Coefficient estimate
`rtaxso`	.0266403
`ltperinc`	.424329
Constant	−.1397201

$n = 48$, $R^2 = .3987$, Root MSE $= .13936$.

Example with $m = 1$ – Price Elasticity of cigarettes

```
. ivreg ltpackpc (ltavgprs = rtaxso )  ltperinc if year == 1995, r first
          Y          X          Z          W

First-stage regressions
```

Source	SS	df	MS		
				Number of obs	= 48
				F(2, 45)	= 14.92
Model	.579443435	2	.289721717	Prob > F	= 0.0000
Residual	.87397801	45	.019421734	R-squared	= 0.3987
				Adj R-squared	= 0.3719
Total	1.45342144	47	.030923861	Root MSE	= .13936

ltavgprs	Coefficient	Std. err.	t	P>|t|	[95% conf.	interval]
ltperinc	.424329	.1198596	3.54	0.001	.1829194	.6657387
rtaxso	.0266403	.0071728	3.71	0.001	.0121937	.041087
_cons	-.1397201	.0433999	-3.22	0.002	-.227132	-.0523083

Example with $m = 1$ – Price Elasticity of cigarettes

Manual second stage: dependent variable `ltpackpc`

Regressor	Coefficient estimate
`hat_ltavgprs`	−1.274014
`ltperinc`	.3865991
Constant	.0012983

$n = 48$, $R^2 = .0900$, Root MSE $= .33569$.

The fitted price comes from the first stage. This gives the 2SLS coefficients; second-stage OLS standard errors do not account for the first stage.

Example with $m = 1$ – Price Elasticity of cigarettes

Dedicated 2SLS: dependent variable `ltpackpc`

Regressor	Coefficient estimate
`ltavgprs`	−1.274014
`ltperinc`	.3865991
Constant	.0012983

$n = 48$, $R^2 = .4479$, Root MSE $= .26147$.

Instrument: `rtaxso`. Exogenous control: `ltperinc`.

A single specialized command gives the 2SLS coefficients and proper 2SLS standard errors, accounting for the first stage.

The General IV Regression Assumptions

Consider the following specification

$$Y_i = \beta_0 + \beta_1 X_{1i} + \ldots \beta_k X_{ki} + \beta_{k+1} W_{1i} + \cdots + \beta_{k+r+1} W_{ri} + u_i$$

and m instruments, $Z_{1i}, \ldots, Z_{mi}$. Under the following assumptions, the 2SLS estimator is consistent and has a sampling distribution that, in large samples, is approximately normal

1. $\mathbb{E}(u_i \mid W_{1i}, \ldots, W_{ri}) = 0$
2. $(Y_i, X_{1i}, \ldots, X_{ki}, W_{1i}, \ldots, W_{ri}, Z_{1i}, \ldots, Z_{mi})$ are i.i.d.
3. Large outliers are unlikely: The X's, W's, Z's, and Y have nonzero finite fourth moments;
4. The instruments $Z_{1i}, \ldots, Z_{mi}$ are valid:
   - 4.1 *Instrument exogeneity:* $corr(Z_{1i}, u_i) = 0, \ldots, corr(Z_{mi}, u_i) = 0$
   - 4.2 *Instrument relevance:* Suppose the second stage regression could be run using the predicted values from the *population* first stage regression. Then there is no perfect multicollinearity in this (infeasible) second-stage regression. Intuitively, the instruments must provide enough information about the exogenous movements in these variables to sort out their separate effects on Y.

Checking Assumption 4.1 – Instrument exogeneity

- Instrument exogeneity: All the instruments are uncorrelated with the error term
- In other words, Z can only impact Y through X
- If the instruments are correlated with the error team, the first stage of 2SLS cannot isolate a component of X that is uncorrelated with the error team, so $\hat{X}$ is correlated with u and 2SLS inconsistent.
- If there are more instruments than endogenous regressors, it is possible to test – partially – for instrument exogeneity.
- Intuition: Consider the simplest case:

$$Y_i = \beta_0 + \beta_1 X_i + u_i$$

Suppose there are two valid instruments: Z_{1i} and Z_{2i}, then you can compute two separate 2SLS estimates. If these 2 2SLS estimates are very different from each other, then something must be wrong: one or the other (or both) of the instruments must be invalid

Checking Assumption 4.1 – Instrument exogeneity

The **J-test of overidentifying restrictions** (Anderson-Rubin test) makes this comparison in a statistically precise way.

Suppose you have $m > k$. Then,

1. Estimate the residuals $\hat{u}_i^{2SLS}$ from

$$Y_i = \beta_0 + \beta_1 X_{1i} + \ldots \beta_k X_{ki} + \beta_{k+1} W_{1i} + \cdots + \beta_{k+r+1} W_{ri} + u_i$$

2. Estimate the following model

$$\hat{u}_i^{2SLS} = \gamma_0 + \gamma_1 Z_{1i} + \ldots \gamma_m Z_{mi} + \gamma_{m+1} W_{1i} + \cdots + \gamma_{m+r+1} W_{ri} + e_i$$

3. Recover the F-statistic testing $H_0 : \gamma_0 = \gamma_1 = \cdots = \gamma_m = 0$. The overidentifying restrictions test statistic is $J = mF$
4. Under the maintained assumptions and the null hypothesis that all the instruments are exogenous, in large samples $J \sim \chi^2_{m-k}$ where $m - k$ is the degree of *over*identification

Checking Assumption 4.2 – Instrument Relevance

In the first stage, we regress

$$X_i = \pi_0 + \pi_1 Z_{1i} + \pi_2 Z_{2i} + \cdots + \pi_m Z_{mi} + \pi_{m+1} W_{1i} + \pi_{m+2} W_{2i} + \cdots + \pi_{m+k} W_{ki} + u_i$$

The instruments are relevant *if at least one* of $\pi_1, \ldots, \pi_m$ is non-zero
The instruments are said to be *weak* if all the $\pi_1, \ldots, \pi_m$ are either zero or nearly zero

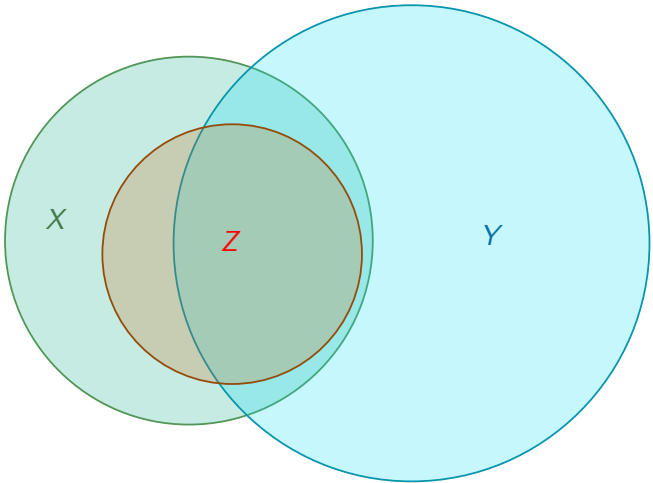
$\Rightarrow$ **Weak instruments** explain very little of the variation in X, beyond that explained by the W's.

If instruments are weak, the sampling distribution of 2SLS and its t-statistic are not (at all) normal, even with n large.

Remember that $\hat{\beta}_1^{2SLS} = \frac{s_{ZY}}{s_{ZX}} \xrightarrow{P} \frac{cov(Z_i, Y_i)}{cov(Z_i, X_i)}$ and when instruments are weak, $cov(Z_i, X_i)$ tends to zero...

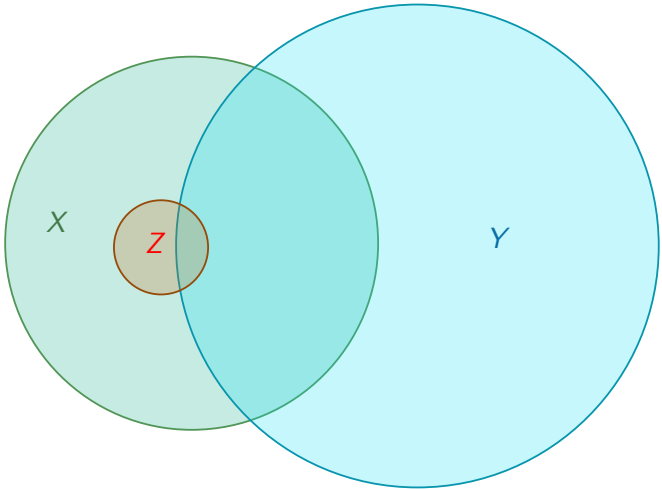
We cannot conduct statistical inference with the usual methods when instruments are weak (inference under weak instruments is beyond the scope of this course).

Checking Assumption 4.2 – Instrument Relevance



Z is a **strong** instrument

Checking Assumption 4.2 – Instrument Relevance



Z is a **weak** instrument

Checking Assumption 4.2 – Instrument Relevance

So... how relevant must the instruments be to draw causal inference in practice? How weak is weak?

- Staiger and Stock (1997)'s Rule of thumb (1 endogenous variable!): Reject that your instruments are weak if $F \geq 10$
- Stock and Yogo (2005) formalized and generalized Staiger and Stock (1997) : Reject that your instruments are weak if $F > J_{10}(m)$ where $J_{10}(m)$ is chosen such that $P(F > J_{10}(m) \mid \text{2SLS bias} \leq 10\% \times \text{OLS bias}) = 0.05$

The literature on weak instrument testing is still ongoing (technical details are beyond the scope of this class)

So if you have weak instruments,

- Get better instruments (often easier said than done!)
- If you have many instruments, some are probably weaker than others and it's a good idea to drop the weaker ones (dropping an irrelevant instrument will increase the first-stage F)
- If you only have a few instruments, and all are weak, then you need to switch from 2SLS to a method that is robust to weak instruments.

Summary

- A valid instrument lets us isolate a part of X that is uncorrelated with u, and that part can be used to estimate the effect of a change in X on Y

- IV regression hinges on having valid instruments:
  1. Exogeneity: Test overidentifying restrictions via the J-statistic + exercise your judgment
  2. Relevance: Check via first-stage F

- A valid instrument isolates variation in X that is "as if" randomly assigned.

- The critical requirement of at least m valid instruments cannot be tested – you must use your head!

Material I

– **Textbooks:**
  - Introduction to Econometrics, 4th Edition, Global Edition, by Stock and Watson – Chapters 12 - 13